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BrightLight Tutorials

Data Analytics

Exercise 1 : SQL Fundamentals

Database : employees_db

Question 01 : Select Statement

- Retrieving all the columns from the employee table

Syntax :

```
SELECT *
From employee_db;
```

Answer :

id	First_name	Last_name	Department	Salary	Hire_date	City
1	John	Doe	IT	55 000	2018-06-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53 000	2021-03-25	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Francisco

QUESTION 02 : SELECT DISTINCT STATEMENT

- Finding all the unique department in the employee table

SYNTAX :

```
SELECT DISTINCT Department
FROM employee_db ;
```

ANSWER :

Department
IT
HR
Finance
Marketing

QUESTION 03 : ORDER BY STATEMENT

- Retrieving all employees First_name and last_name order by Salary in descending order

SYNTAX :

```
SELECT First_name, last_name, salary
FROM employee_db
ORDER BY Salary DESC ;
```

ANSWER : IN THE NEXT PAGE

ANSWER : QUESTION 03

First_name	Last_name	Salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000
Daniel	Clark	53 000
David	White	52 000
Jessica	Moore	51 000
Laura	Hall	50 000
Jane	Smith	48 000

QUESTION 04 : LIMIT STATEMENT

- Retrieving top 5 highest-paid employees

SYNTAX :

```
SELECT  First_name, last_name, salary
From    employees_db
Limit   5 ;
```

ANSWER :

first_name	Last_name	Salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000

QUESTION 05 : WHERE STATEMENT

- Find employee who work in the IT Department

SYNTAX :

```
SELECT first_name, last_name, Department
FROM employee_db
WHERE Department = 'IT' ;
```

ANSWER :

First_name	Last_name	Department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

QUESTION 06 : AND STATEMENT

- Find employee who work in the HR Department and have salary greater than 58 000

SYNTAX :

```
SELECT First_name, Last_name, Department, salary
FROM employee_db
WHERE Department = 'Finance' AND Salary > 58 000 ;
```

ANSWER :

First_name	Last_name	Department	salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000

QUESTION 07 : OR STATEMENT

- find employees who work in the HR Department OR the Marketing department.

SYNTAX:

```
SELECT First_name, Department
FROM employee_db
WHERE department = 'HR' OR Department = 'Marketing';
```

ANSWER:

First_name	Department
Jane	HR
David	Marketing
Jessica	HR
Daniel	Marketing

QUESTION 08 : NOT STATEMENT

- Finding employees who do not work in IT

SYNTAX:

```
SELECT First_name, Department
FROM employee_db
WHERE Department =! 'IT';
```

ANSWER:

First_name	Department
Jane	HR
Mike	Finance
David	Marketing
Robert	Finance
Jessica	HR
Daniel	Marketing

QUESTION 09: IN STATEMENT

- Finding employees who are in the HR, IT and Finance department

SYNTAX:

```
SELECT  First_name, department
FROM    employee_db
WHERE   department IN ('HR', 'IT' or 'Finance')
```

ANSWER:

First_name	Department
John	IT
Jane	HR
Mike	Finance
Sarah	IT
Emily	IT
Robert	Finance
Jessica	HR
Laura	IT

QUESTION 10: COMBINING CONDITION

- Finding employee in IT and earning greater than 50 000 and are in New YORK

SYNTAX

```
SELECT  First_name, Salary, City, department
FROM    employee_db
WHERE   Department = 'IT' Salary > 50000 AND City = NY
```

QUESTION 10 : ANSWER

First_name	Department	City	Salary
John	IT	New York	55 000
Sarah	IT	New York	53 000

QUESTION 11 : COMBINING WHERE, AND, and ORDER BY

SYNTAX :

```
SELECT First_name, Last_name, Department, Salary
FROM employee_db
WHERE department = ('Finance' or 'Marketing') AND Salary
                    > 52 000
ORDER BY Salary DESC;
```

ANSWER :

First_name	Last_name	Department	Salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	53 000